

REGEX FOUNDATION (From Basics)

1. How to extract digits

Example input

123ABc

Regex

[0-9]

Explanation

[0-9] matches any single digit from 0 to 9

It does not match alphabets or special characters

2. How to extract alphabets

Example input

123ABc

Regex

[A-Za-z]

Explanation

[A-Z] → only uppercase letters

[a-z] → only lowercase letters

[A-Za-z] → both uppercase and lowercase letters

3. ^ (start of string)

Explanation

^ checks whether the pattern exists at the beginning of the string

If the string does not start with the pattern, the result is empty

Regex

^[0-9]

Examples

123abc → 1

abc123 → no output

This means the string must start with a digit.

4.\$ (end of string)

Explanation

\$ checks whether the pattern exists at the end of the string

If the string does not end with the pattern, the result is empty

Regex

[0-9]\$

Examples

123abc → no output

abc123 → 3

This means the string must end with a digit.

5. Extracting exact number of characters using { }

Extract exactly 2 digits

Regex

[0-9]{2}

Input

12345678abc

Output

12

Extract exactly 2 alphabets

Regex

[a-zA-Z]{2}

Input

abcdefghi123

Output

ab

Note

{n} means exactly n characters.

6. + (one or more characters)

Explanation

+ is a reserved character in regex

It means one or more continuous matches

Digits example

Regex

[0-9]+

123abc → 123

abc123 → 123

Alphabets example

Regex

[a-zA-Z]+

abc123 → abc

Important point

+ does not extract a single character

It extracts all continuous matching characters

7. Principle of continuity

Explanation

Regex always matches continuous characters only

It stops when the pattern breaks

Example 1

Input

123abc567

Regex

[0-9]+

Output

123

It does not return 123567 because the digits are not continuous.

Example 2

Input

abc123cde

Regex

[a-zA-Z]+

Output

abc

7. (a) Extract everything (digits + alphabets)

Regex

[0-9a-zA-Z]+

Input

123abc567

Output

123abc567

(b) Extract digits from the middle

Regex

[0-9]+

Input

abc123gef

Output

123

8. Extracting data from the middle (Email example)

Input

karthik@gmail.com

Extract domain part

Regex

@[\a-zA-Z.]+\a

Output

@gmail.com

Extract top-level domain

Regex

\\.[a-zA-Z]+\$

Output

.com

9. Reserved characters in regex

Some characters have special meaning and must be escaped using \.

Common reserved characters:

+

.

^

\$

Phone number example

Input

+91-9989454737

Regex

\+91-[0-9]{10}

10. Removing unwanted characters using REPLACE in MySQL

If @ appears in the output and you want to remove it:

REPLACE(

REGEXP_SUBSTR(email, '@[a-zA-Z.]+'),

'@',

"

)

Output

gmail.com

11. Email – Regex, DQ and Data Validation

When working with emails, we usually do two things:

Data Quality (DQ) check

Data Validation (whether the email format is correct)

Valid email examples

karthik@gmail.com

karthik44@gmail.com

karthik.kumar@gmail.com

karthik_kumar44@gmail.com

karthik-kumar44@gmail.com

fardeen@st2.ai

karthik@sb2.com

vinay@infosys.com

Invalid / unwanted part example

@gmail.com

Here, username is missing.

We want only:

gmail.com

Email structure (very important)

An email has three logical parts:

username @ domain . extension

Example:

karthik.kumar44@gmail.com

A) Before @ (username part)

Allowed characters:

a-z

A-Z

0-9

dot (.)

underscore (_)

hyphen (-)

So valid usernames are:

karthik

karthik44

karthik.kumar

karthik_kumar44

karthik-kumar44

Regex part:

[a-zA-Z0-9._-]+

B) After @ (domain name)

Allowed characters:

a–z

A–Z

0–9

Examples:

gmail

infosys

sb2

st2

Regex part:

[a-zA-Z0-9]+

C) After . (domain extension)

Common extensions:

.com

.org

.ai

Length rule:

Usually 2 or 3 characters

Examples:

com

org

ai

Regex part:

[a-zA-Z]{2,3}

Final combined email validation regex

Putting all parts together:

```
[a-zA-Z0-9._-]+@[a-zA-Z0-9]+\.[a-zA-Z]{2,3}
```